

AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH

(AIUB)

FACULTY OF SCIENCE & TECHNOLOGY

INTRODUCTION TO DATBASE

Spring 2023-2024

Section: J, Group: 05

PROJECT NAME

Fund Raiser Management System

Submitted By:

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1. Introduction :

Introducing the Fundraiser Management System (FMS), an innovative solution tailored to streamline the complexities of organizing and executing fundraising campaigns. FMS serves as the cornerstone for individuals, charities, and organizations alike, offering a comprehensive suite of tools to orchestrate successful fundraising endeavors. With its intuitive interface and robust features, FMS simplifies campaign management, empowering users to set goals, define strategies, and track progress with ease. From DONER management and online fundraising to event planning and reporting, FMS provides a centralized platform to efficiently coordinate every aspect of the fundraising process. By harnessing the power of data analytics and engagement tools, FMS enables users to cultivate meaningful relationships with DONERs, optimize resource allocation, and measure the impact of their initiatives. With FMS, organizations can navigate the intricacies of fundraising with confidence, driving greater efficiency, effectiveness, and ultimately, making a tangible difference in their communities.

2. Scenario Description

StudentID1: 23-50860-1 Name: 1. RISA,SUMIYA HAIDER	StudentID3: 23-51046-1 Name: MST. SHARAF ANIKA ESHA
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StudentID5: 23-51079-1 Name: ALIF , MD RAISUL HOQUE CHOWDHURY	
CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study	
PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.	Marks

The database for the Fund Raiser Management System includes entities such as Events, DONERS, Donations, Organizations and Fundraising team. These represent the fundraising events organized by various organizations. Each event has attributes such as E_ID (Primary key), E_Name, E_Date, E_Location, and E_Type. Individuals or entities who contribute funds to support the fundraising activities. DONERs are characterized by attributes like DONER_ID(Primary key), DONER_Name, DONER_Email, and DONER_Phone. Records the donations made by DONERs to specific events. It includes attributes such as D_ID(Primary key), D_Amount, and D_Date. Entities that host or sponsor fundraising events. Each organization is described by attributes such as Org_ID(Primary key), Org_Name, Org_Address, and Org_Contact. In many to one

relationship a fundraising team includes several events, an event is associated with at most one fundraiser team. In one-to-many relationships, an event is associated with several organization, an organization is associated with at the most one event. An event can receive many donations via doners. There is a ternary relationship among Organization, Donation and Doner. An Organization can have many donation via doners and Many doners can donate in an organization.

3. ER Diagram:

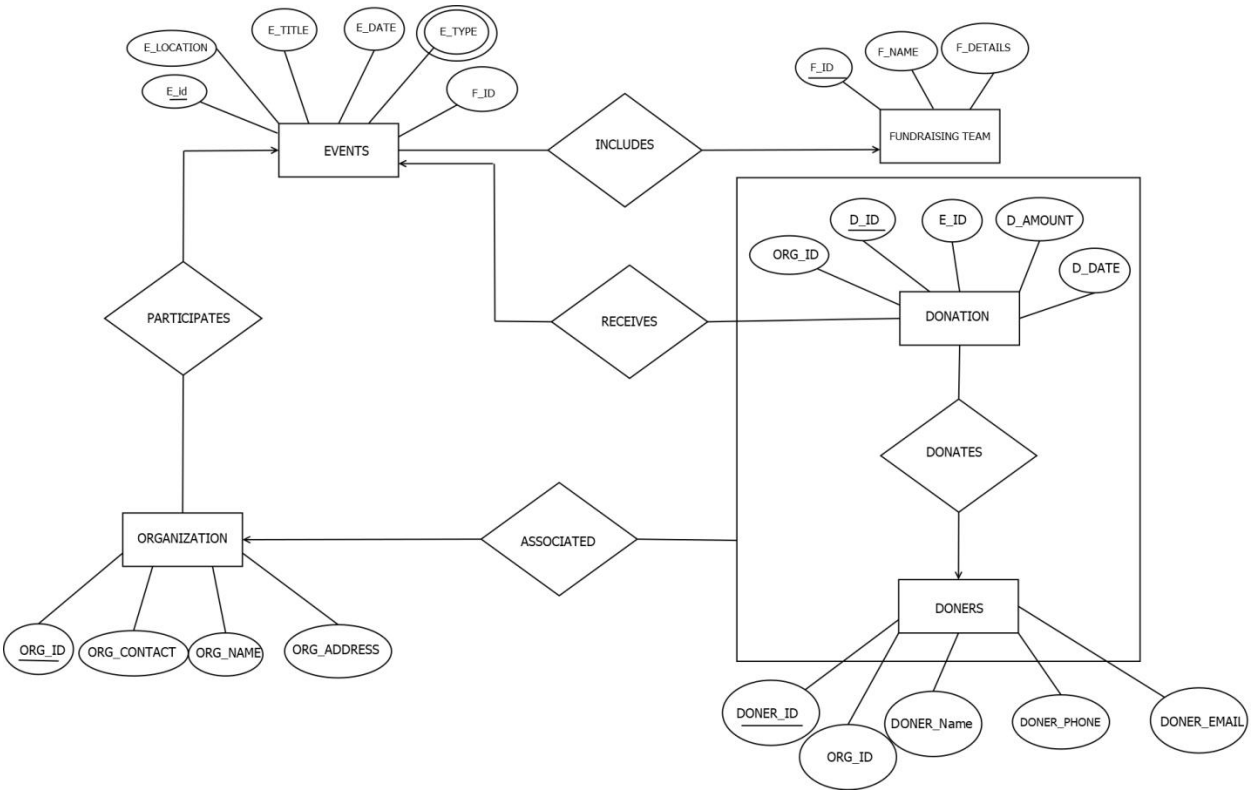


Figure: ER DIAGRAM

4. NORMALIZATION:

4.1 INCLUDES RELATION:

UNF:

E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID, F_NAME, F_DETAILS.

1NF:

E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID, F_NAME, F_DETAILS.

2NF:

1. E_ID(PK), E_LOCATION, E_TITLE, E_DATE, E_TYPE.
2. F_ID, F_NAME, F_DETAILS.
3. F_ID(PK), E_ID(FK)

3NF:

1. E_ID(PK), E_LOCATION, E_TITLE, E_DATE, E_TYPE.
2. F_ID, F_NAME, F_DETAILS.
3. F_ID(PK), E_ID(FK)

4.2 PARTICIPATES RELATION:

UNF:

1. E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID, ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS

1NF:

1. E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID, ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS
2. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS
3. E_ID(PK), ORG_ID(FK)

2NF:

1. E_ID(PK), E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID
2. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

3. E_ID(PK), ORG_ID(FK)

3NF:

1. E_ID(PK), E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID

2. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

3. E_ID(PK), ORG_ID(FK)

4.3 RECEIVES RELATION:

UNF:

1. E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID, D_ID, D_AMOUNT, D_DATE, ORG_ID.

1NF:

1. E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID, D_ID, D_AMOUNT, D_DATE, ORG_ID.

2. D_ID(PK), D_AMOUNT, D_DATE, ORG_ID

3. E_ID(PK), D_ID(FK)

2NF:

1. E_ID(PK), E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID

2. D_ID(PK), D_AMOUNT, D_DATE, ORG_ID

3. E_ID(PK), D_ID(FK)

3NF:

1. E_ID(PK), E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID

2. D_ID(PK), D_AMOUNT, D_DATE, ORG_ID

3. E_ID(PK), D_ID(FK)

4.4 DONATES RELATION:

UNF:

D_ID(PK), D_AMOUNT, D_DATE, ORG_ID, DONERS_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL.

1NF:

1. D_ID(PK), D_AMOUNT, D_DATE, ORG_ID, DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL.

2. DONER_ID(PK), DONER_PHONE, DONER_NAME, DONER_EMAIL.

3. D_ID(PK), DONER_ID(FK)

2NF:

1. D_ID(PK), D_AMOUNT, D_DATE, ORG_ID.

2. DONER_ID(PK), DONER_PHONE, DONER_NAME, DONER_EMAIL.

3. D_ID(PK), DONER_ID(FK)

3NF:

1. D_ID(PK), D_AMOUNT, D_DATE, ORG_ID.

2. DONER_ID(PK), DONER_PHONE, DONER_NAME, DONER_EMAIL.

3. D_ID(PK), DONER_ID(FK)

4.5 ASSOCIATED RELATION:

UNF:

ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS, D_ID, D_AMOUNT, D_DATE

1NF:

1. ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS, D_ID, D_AMOUNT, D_DATE

2. D_ID(PK), D_AMOUNT, D_DATE, ORG_ID

3. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

D_ID(PK), ORG_ID(FK)

2NF:

1. D_ID(PK), D_AMOUNT, ORG_ID, D_DATE

2. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

3. D_ID(PK), ORG_ID(FK)

3NF:

1. D_ID(PK), D_AMOUNT, ORG_ID, D_DATE

2. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

3. D_ID(PK), ORG_ID(FK)

4.6 ASSOCIATED RELATION:

UNF:

ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS, DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL

1NF:

1.ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS, DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL

2.DONER_ID(PK), DONER_PHONE, DONER_NAME, DONER_EMAIL

3. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

DONER_ID(PK), ORG_ID(PK)

2NF:

1. DONER_ID(PK), DONER_PHONE, DONER_NAME, DONER_EMAIL

2. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

3. DONER_ID(PK), ORG_ID(PK)

3NF:

1. DONER_ID(PK), DONER_PHONE, DONER_NAME, DONER_EMAIL

2. ORG_ID(PK), ORG_CONTACT, ORG_NAME, ORG_ADDRESS

3. DONER_ID(PK), ORG_ID(PK)

FINALIZATION:

FINAL TABLE:

1. E_ID (PK), E_LOCATION, E_NAME, E_TITLE, E_DATE, E_TYPE, F_ID

2. F_ID (PK), F_NAME, F_DETAILS

3. F_ID (PK), E_ID (FK)

4. ORG_ID (PK), ORG_NAME, ORG_CONTACT, ORG_ADDRESS

5. E_ID (PK), ORG_ID (FK)

6. D_ID (PK), D_AMOUNT, D_DATE, ORG_ID

7. E_ID (PK), D_ID (FK)

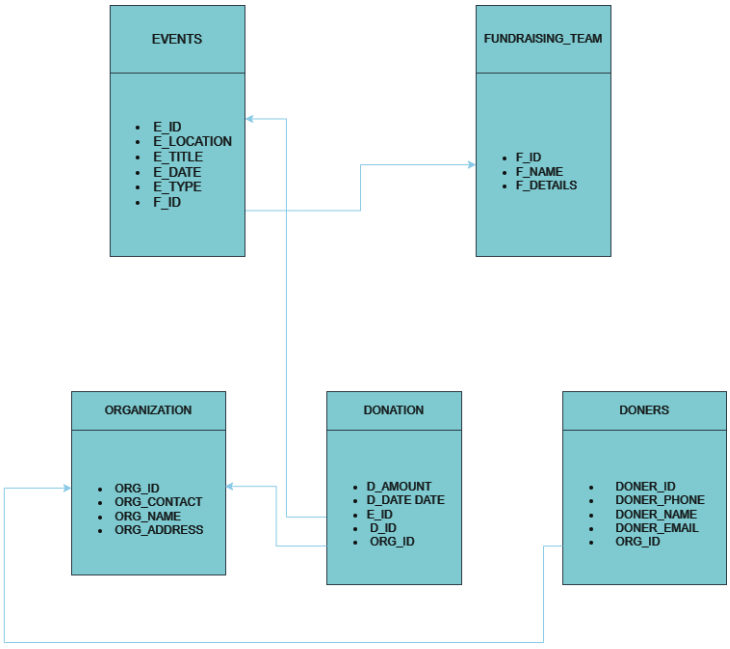
8. DONER_ID (PK), DONER_NAME, DONER_PHONE, DONER_EMAIL

9. D_ID (PK), DONER_ID (FK)

10. D_ID (PK), ORG_ID (FK)

11. DONER_ID (PK), ORG_ID (FK)

Schema Diagram:

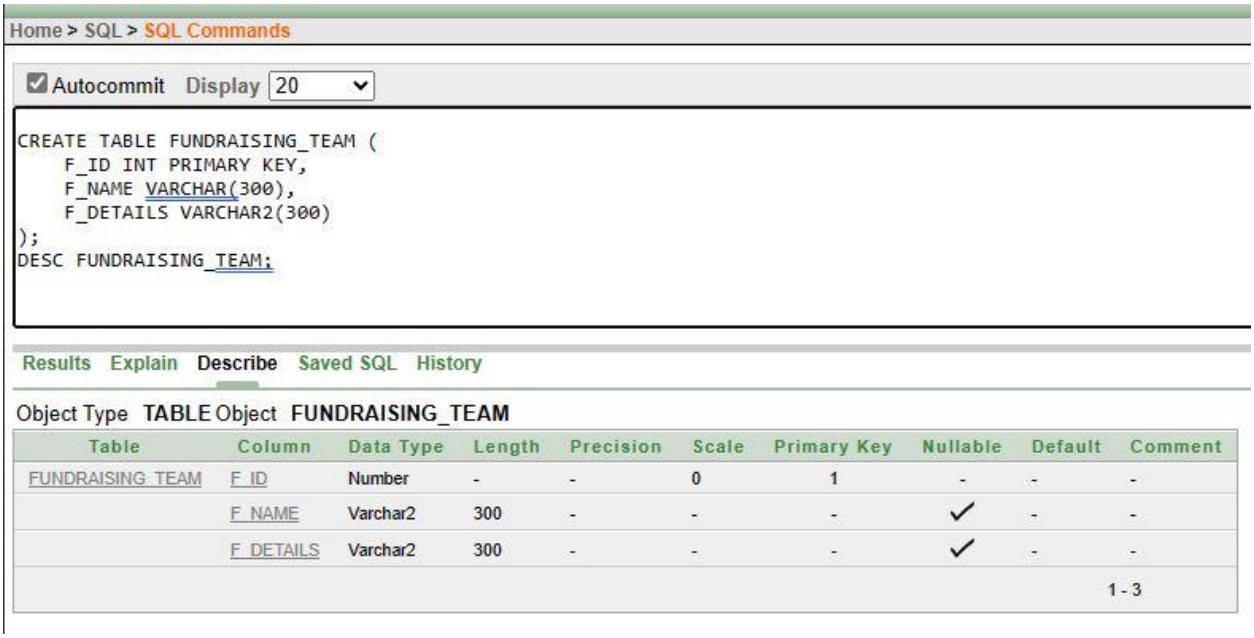


6. Table Creation

StudentID1: 23-50860-1 Name: 1. RISA,SUMIYA HAIDER	StudentID3: 23-51046-1 Name: MST. SHARAF ANIKA ESHA
StudentID2: 23-51050-1 Name: OVI ,AYON KUMAR BHOWMICK	StudentID4: 23-51096-1 Name: AHAMMED, MD. SHAPARAN
StudentID5: 23-51079-1 Name: ALIF , MD RAISUL HOQUE CHOWDHURY	
CO4: Creating DML, DDL using Oracle and connection with ODBC/JDBC for existing JAVA application	
PO-e-2: Use modern engineering and IT tools for prediction and modeling of complex computer science and engineering problem	Marks

6.1 FUNDRAISING_TEAM Table:

```
1. CREATE TABLE FUNDRAISING_TEAM (
    F_ID INT PRIMARY KEY,
    F_NAME VARCHAR(300),
    F_DETAILS VARCHAR2(300)
);
DESC FUNDRAISING_TEAM;
```



6.2 EVENTS Table:

1. CREATE TABLE EVENTS (

E_ID INT PRIMARY KEY,

E_LOCATION VARCHAR(260),

E_TITLE VARCHAR(260),

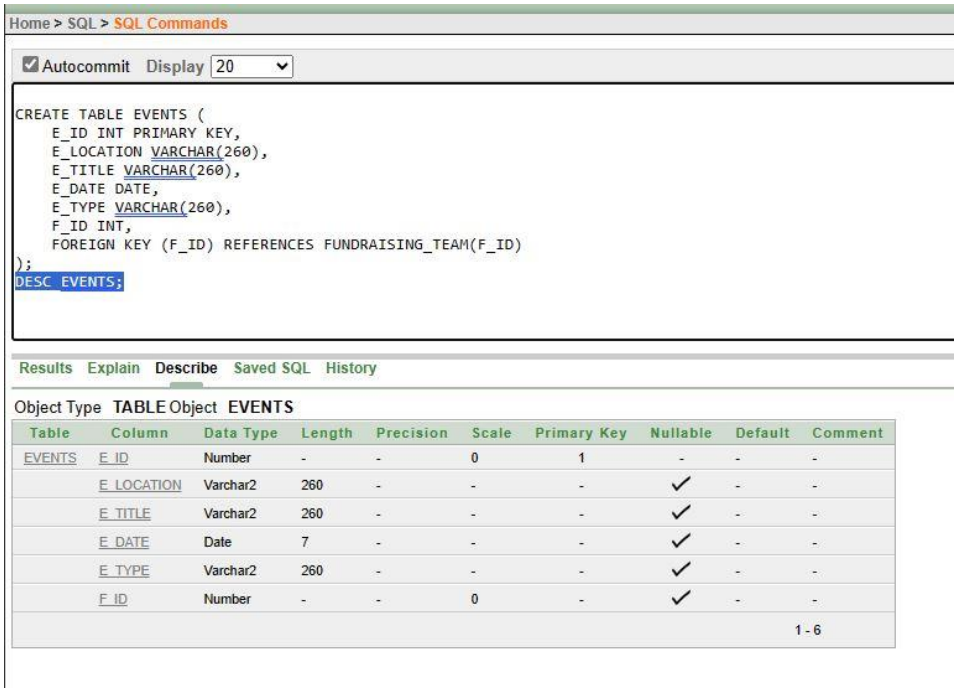
E_DATE DATE,

E_TYPE VARCHAR(260),

F_ID INT,

FOREIGN KEY (F_ID) REFERENCES FUNDRAISING_TEAM(F_ID));

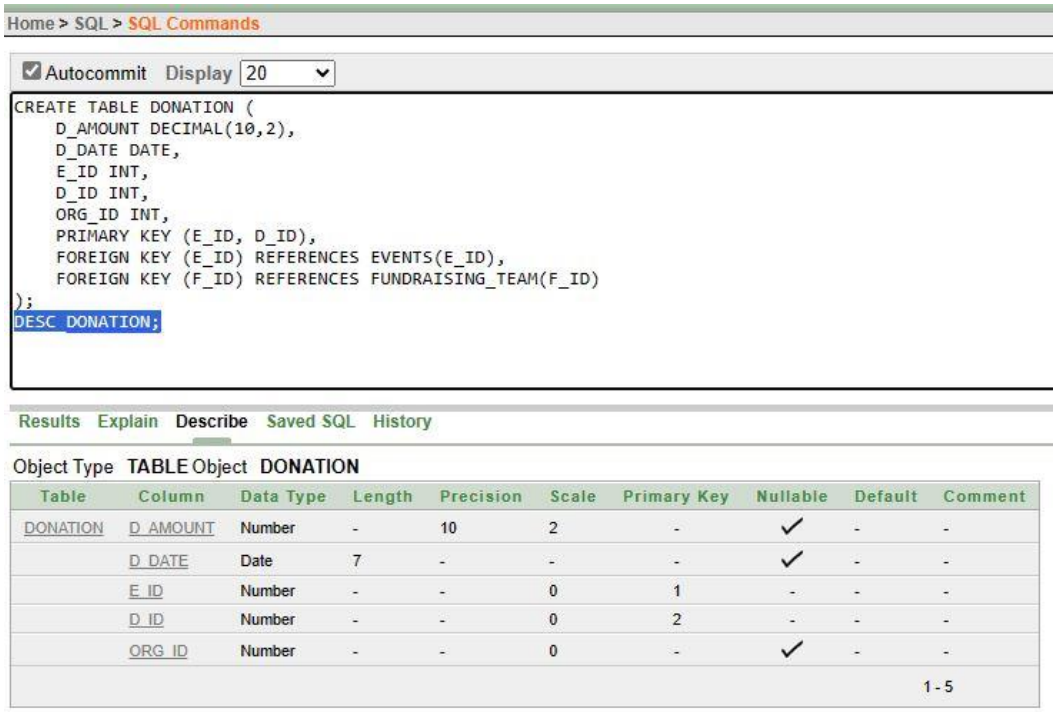
DESC EVENTS;



6.3 DONATION Table:

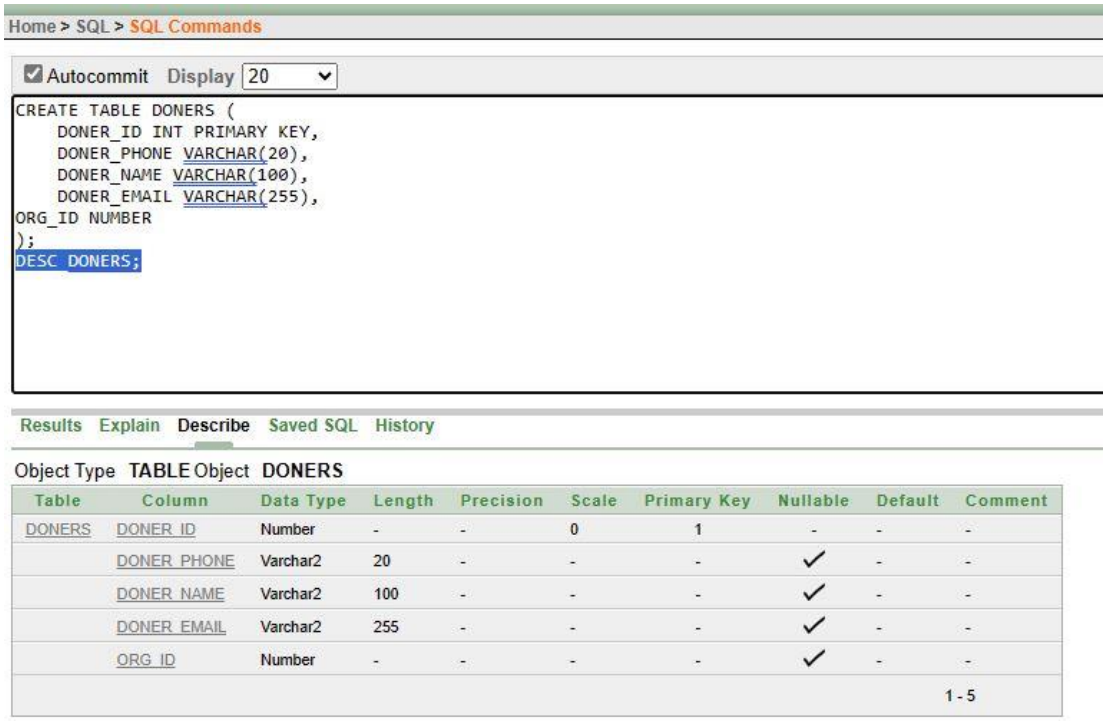
3.CREATE TABLE DONATION (
D_AMOUNT DECIMAL(10,2),
D_DATE DATE,
E_ID INT,
D_ID INT,

```
    ORG_ID INT,
    PRIMARY KEY (E_ID, D_ID),
    FOREIGN KEY (E_ID) REFERENCES EVENTS(E_ID),
    FOREIGN KEY (F_ID) REFERENCES FUNDRAISING_TEAM(F_ID)
);
DESC DONATION;
```



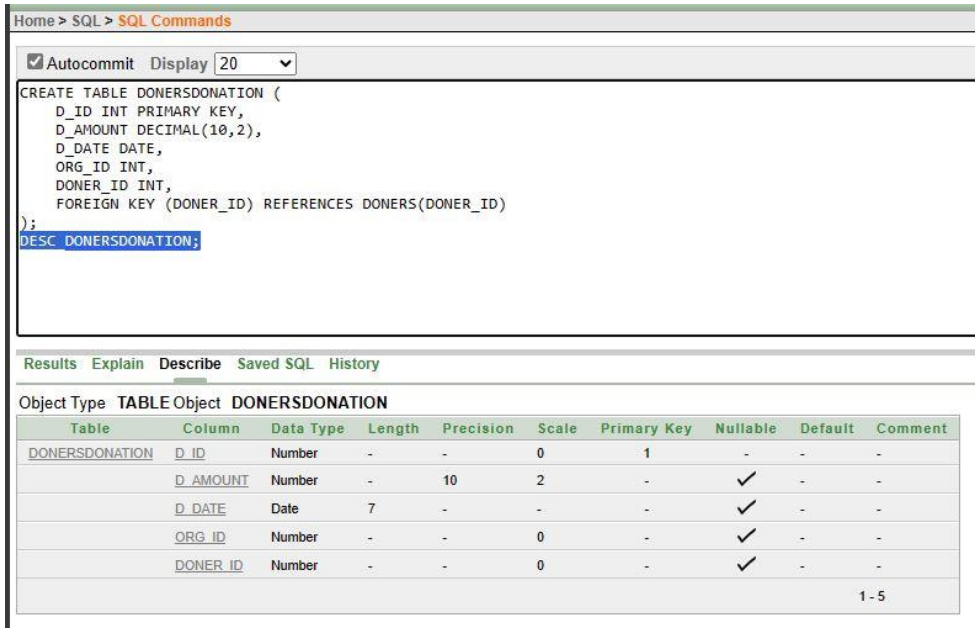
6.4 DONERS TABLE:

```
4. CREATE TABLE DONERS (
    DONER_ID INT PRIMARY KEY,
    DONER_PHONE VARCHAR(20),
    DONER_NAME VARCHAR(100),
    DONER_EMAIL VARCHAR(255),
ORG_ID NUMBER
);
DESC DONERS;
```



6.5 DONERSDONATION TABLE:

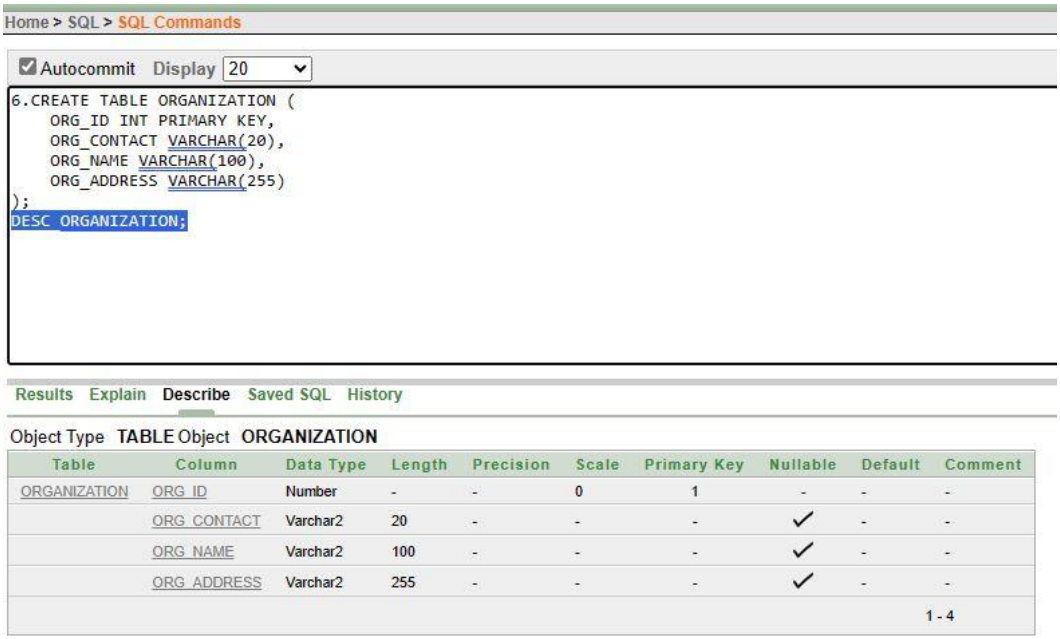
```
5. CREATE TABLE DONERSDONATION (
    D_ID INT PRIMARY KEY,
    D_AMOUNT DECIMAL(10,2),
    D_DATE DATE,
    ORG_ID INT,
    DONER_ID INT,
    FOREIGN KEY (DONER_ID) REFERENCES DONERS(DONER_ID)
);
DESC DONERSDONATION;
```



6.6 ORGANIZATION TABLE:

6.CREATE TABLE ORGANIZATION (

ORG_ID INT PRIMARY KEY,
ORG_CONTACT VARCHAR(20),
ORG_NAME VARCHAR(100),
ORG_ADDRESS VARCHAR(255)
);
DESC ORGANIZATION;



7 INSERTING VALUES:

7.1 Insert into FUNDRAISING_TEAM Table:

1.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (1, 'FUNDRAISING TEAM A', 'RESPONSIBLE FOR CHARITY ORGANIZING EVENTS IN URBAN AREAS.');

2.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (2, 'FUNDRAISING TEAM B', 'FOCUSES ON ONLINE FUNDRAISING CAMPAIGNS FOR MEDICAL RESEARCH.');

3.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (3, 'FUNDRAISING TEAM C', 'SPECIALIZES IN FUNDRAISING EVENTS FOR EDUCATIONAL SCHOLARSHIPS.');

4.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (4, 'FUNDRAISING TEAM D', 'SUPPORTS LOCAL NGOS IN ENVIRONMENTAL CONSERVATION PROJECTS.');

5.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (5, 'FUNDRAISING TEAM E', 'ORGANIZES FUNDRAISING MARATHONS FOR HUMANITARIAN AID.');

6.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (6, 'FUNDRAISING TEAM F', 'MANAGES CHARITY AUCTIONS TO RAISE FUNDS FOR COMMUNITY CENTERS.');

7.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (7, 'FUNDRAISING TEAM G', 'COORDINATES FUNDRAISING DINNERS FOR HOMELESS SHELTERS.');

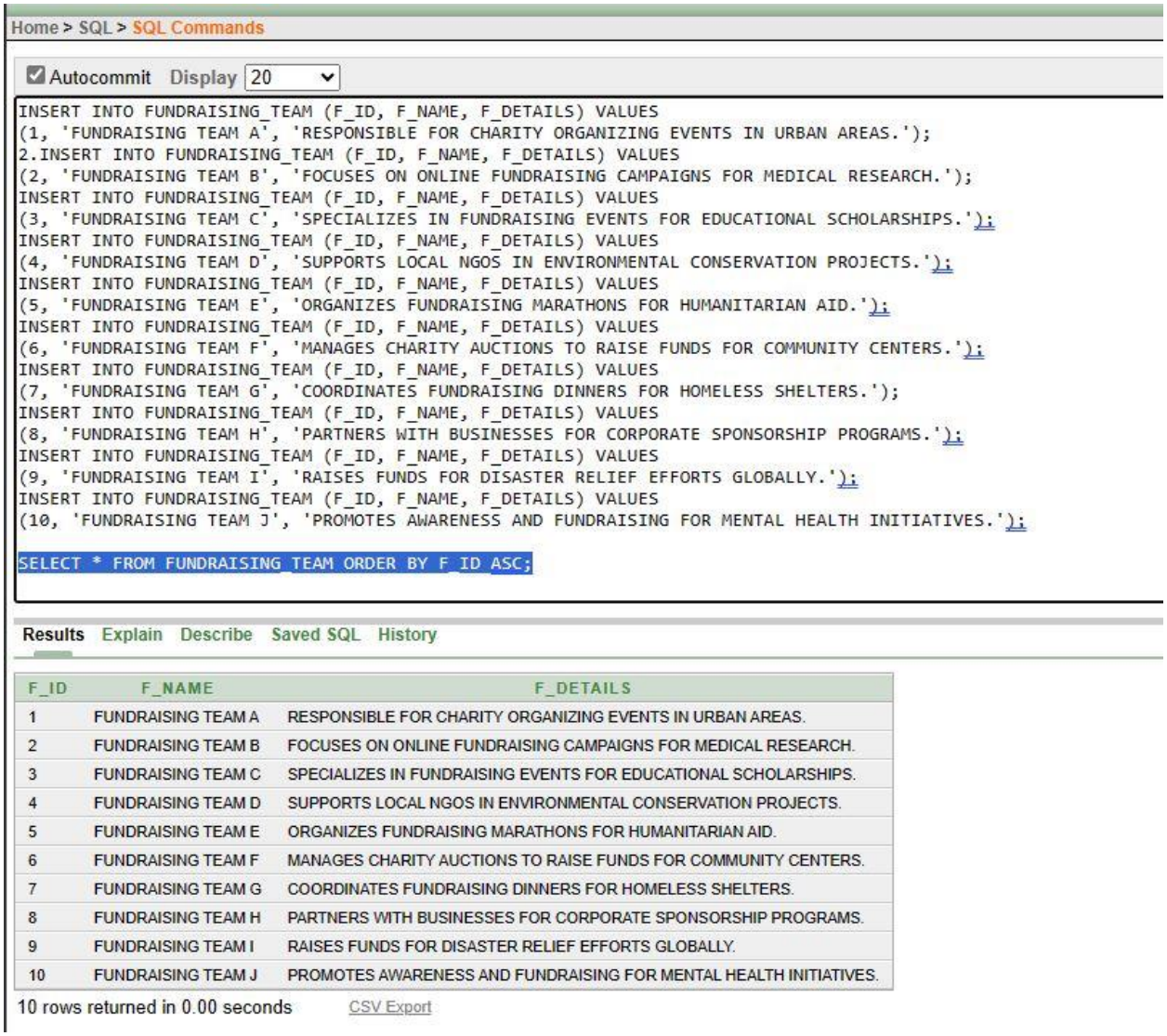
8.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (8, 'FUNDRAISING TEAM H', 'PARTNERS WITH BUSINESSES FOR CORPORATE SPONSORSHIP PROGRAMS.');

9.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES (9, 'FUNDRAISING TEAM I', 'RAISES FUNDS FOR DISASTER RELIEF EFFORTS GLOBALLY.');

10.INSERT INTO FUNDRAISING_TEAM (F_ID, F_NAME, F_DETAILS) VALUES

(10, 'FUNDRAISING TEAM J', 'PROMOTES AWARENESS AND FUNDRAISING FOR MENTAL HEALTH INITIATIVES.');

SELECT * FROM FUNDRAISING_TEAM ORDER BY F_ID ASC;



7.2 Insert into EVENTS Table:

1.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (1, 'RAMPURA,DHAKA', 'CHARITY ', TO_DATE('2024-02-15', 'YYYY-MM-DD'), 'CHARITY', 1);

2.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (2, 'BASHUNDHARA ,DHAKA', ' MEDICAL CAMPAIGN', TO_DATE('2024-02-16', 'YYYY-MM-DD'), 'EVENTS FOR MEDICAL CAMPAIGN', 2);

3.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (3, 'KURIL,DHAKA', ' EDUCATIONAL SCHOLARSHIPS', TO_DATE('2024-02-17', 'YYYY-MM-DD'), 'EVENTS FOR EDUCATIONAL SCHOLARSHIPS', 3);

4.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (4, 'GAZIPUR,DHAKA', ' ENVIRONMENTAL CONSERVATION PROJECT', TO_DATE('2024-02-18', 'YYYY-MM-DD'), ' EVENTS FOR ENVIRONMENTAL CONSERVATION PROJECTS', 4);

5.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (5, 'NORDA,DHAKA', ' MARATHONS FOR HUMANITARIAN AID', TO_DATE('2024-02-19', 'YYYY-MM-DD'), ' EVENTS FOR MARATHONS FOR HUMANITARIAN AID', 5);

6.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (6, 'BANASREE,DHAKA', ' COMMUNITY CENTERS FUNDRAISING', TO_DATE('2024-02-20', 'YYYY-MM-DD'), ' EVENTS FOR FUNDS COMMUNITY CENTERS', 6);

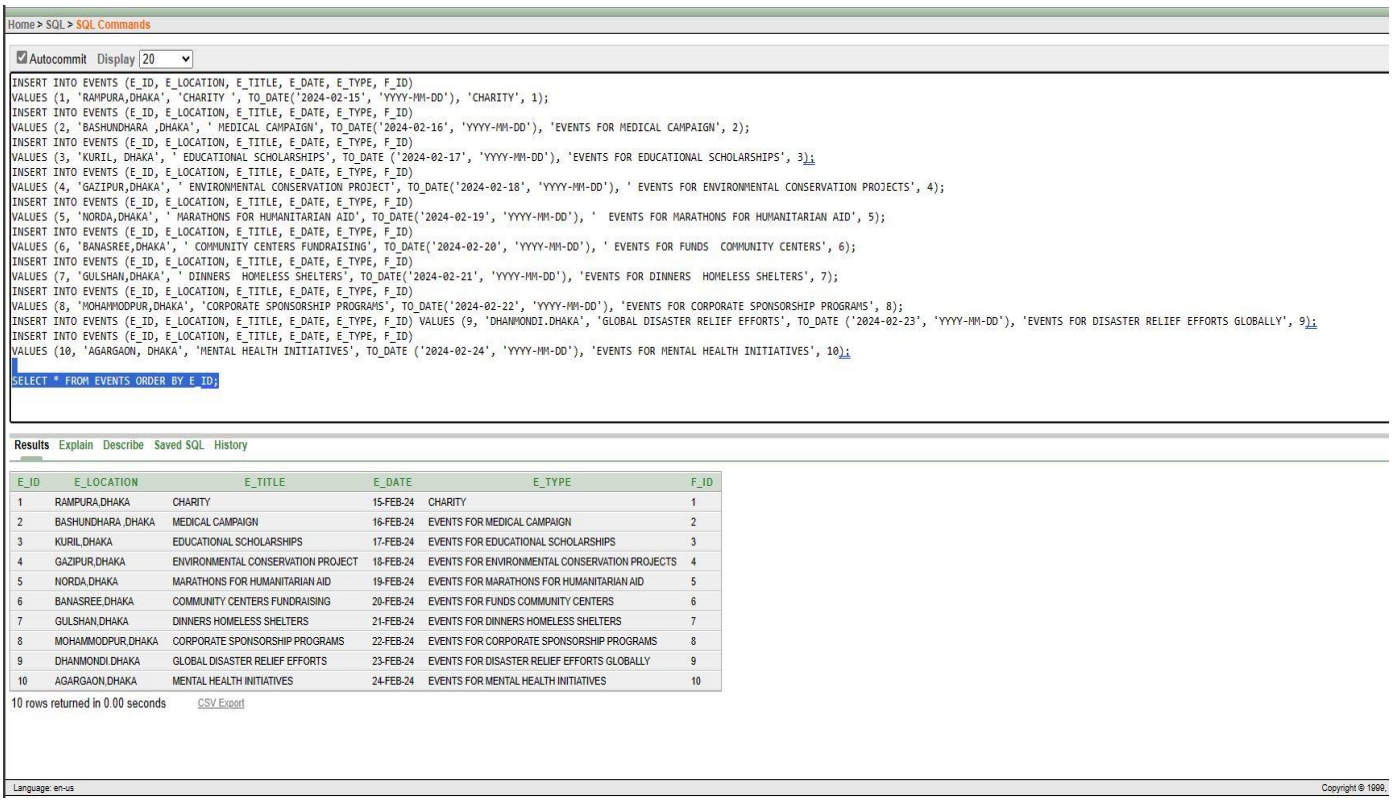
7.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (7, 'GULSHAN,DHAKA', ' DINNERS HOMELESS SHELTERS', TO_DATE('2024-02-21', 'YYYY-MM-DD'), 'EVENTS FOR DINNERS HOMELESS SHELTERS', 7);

8.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (8, 'MOHAMMODPUR,DHAKA', 'CORPORATE SPONSORSHIP PROGRAMS', TO_DATE('2024-02-22', 'YYYY-MM-DD'), 'EVENTS FOR CORPORATE SPONSORSHIP PROGRAMS', 8);

9.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (9, 'DHANMONDI.DHAKA', 'GLOBAL DISASTER RELIEF EFFORTS', TO_DATE('2024-02-23', 'YYYY-MM-DD'), 'EVENTS FOR DISASTER RELIEF EFFORTS GLOBALLY', 9);

10.INSERT INTO EVENTS (E_ID, E_LOCATION, E_TITLE, E_DATE, E_TYPE, F_ID) VALUES (10, 'AGARGAON,DHAKA', 'MENTAL HEALTH INITIATIVES', TO_DATE('2024-02-24', 'YYYY-MM-DD'), 'EVENTS FOR MENTAL HEALTH INITIATIVES', 10);

SELECT * FROM EVENTS ORDER BY E_ID;



7.3 Insert into DONATION Table:

1.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID)
VALUES(10000, TO_DATE ('2024-02-15', 'YYYY-MM-DD'), 1, 1, 1);

2.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID)
VALUES(20000, TO_DATE('2023-01-20', 'YYYY-MM-DD'), 2, 2, 2);

3.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID)
VALUES(30000, TO_DATE('2022-02-28', 'YYYY-MM-DD'), 3, 3, 3);

4.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID)
VALUES(40000, TO_DATE('2021-05-15', 'YYYY-MM-DD'), 4, 4, 4);

5.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID)
VALUES(50000, TO_DATE('2020-02-28', 'YYYY-MM-DD'), 5, 5, 5);

6.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID)
VALUES(60000, TO_DATE('2024-03-11', 'YYYY-MM-DD'), 6, 6, 6);

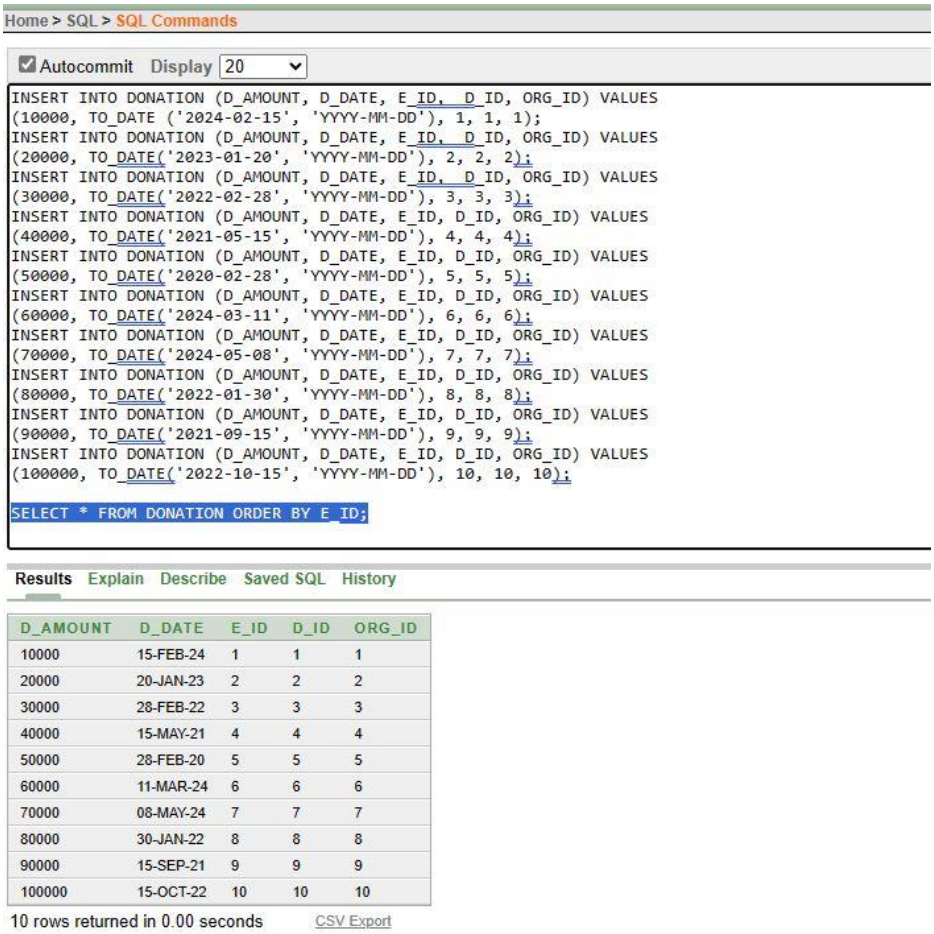
7.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID) VALUES(70000, TO_DATE('2024-05-08', 'YYYY-MM-DD'), 7, 7, 7);

8.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID) VALUES(80000, TO_DATE('2022-01-30', 'YYYY-MM-DD'), 8, 8, 8);

9.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID) VALUES(90000, TO_DATE('2021-09-15', 'YYYY-MM-DD'), 9, 9, 9);

10.INSERT INTO DONATION (D_AMOUNT, D_DATE, E_ID, D_ID, ORG_ID) VALUES(100000, TO_DATE('2022-10-15', 'YYYY-MM-DD'), 10, 10, 10);

SELECT * FROM DONATION ORDER BY E_ID;



7.4 Insert Into DONERS Table:

1.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (1, '01781456541', 'Mohammad Rahman', 'rahman@gmail.com',1);

2.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (2, '01781456542', 'Ali Ahmed', 'ali@gmail.com',2);

3.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (3, '01781456543', 'Sabina Akhtar', 'sabina@gmail.com',3);

4.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (4, '01781456544', 'Mahbub Alam', 'mahbub@gmail.com',4);

5.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (5, '01781456545', 'Jahanara Khanam', 'jahanara@gmail.com',5);

6.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (6, '01781456546', 'Ariful Islam', 'ariful@gmail.com',6);

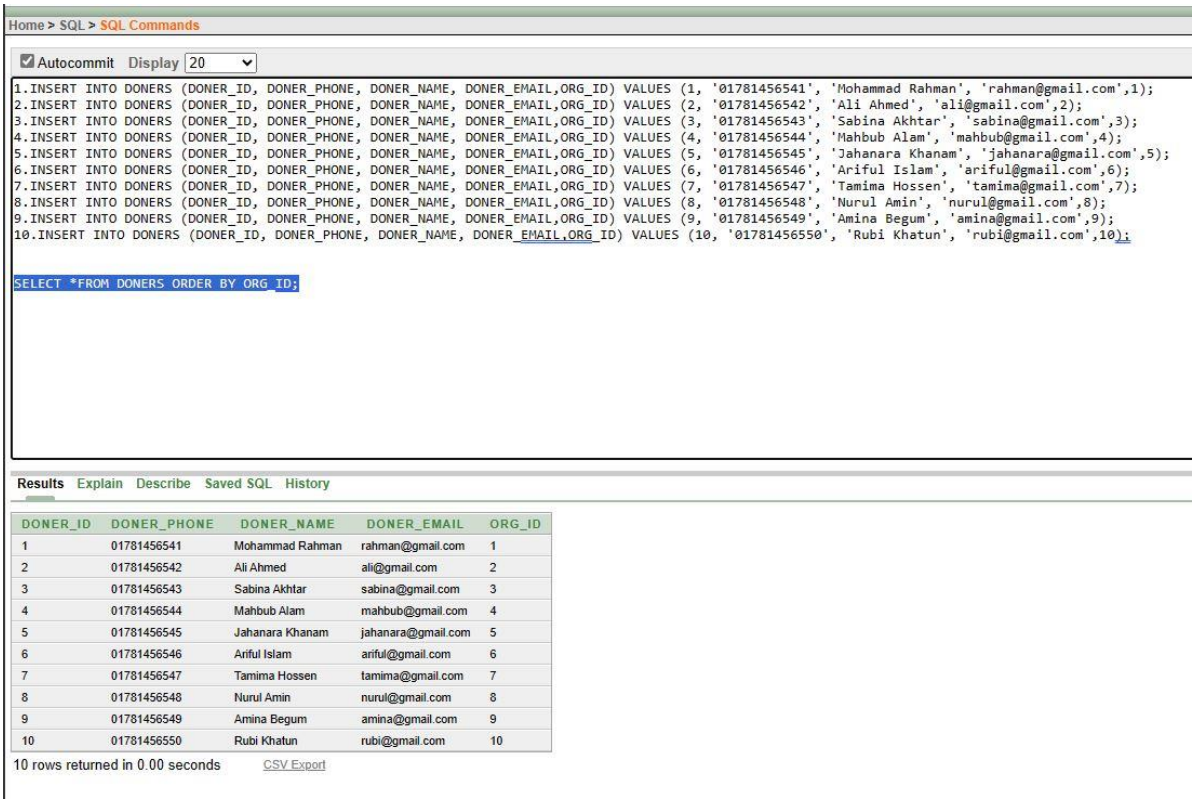
7.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (7, '01781456547', 'Tamima Hossen', 'tamima@gmail.com',7);

8.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (8, '01781456548', 'Nurul Amin', 'nurul@gmail.com',8);

9.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (9, '01781456549', 'Amina Begum', 'amina@gmail.com',9);

10.INSERT INTO DONERS (DONER_ID, DONER_PHONE, DONER_NAME, DONER_EMAIL,ORG_ID) VALUES (10, '01781456550', 'Rubi Khatun', 'rubi@gmail.com',10);

SELECT *FROM DONERS ORDER BY ORG_ID;



7.5 Insert Into DONERSDONATION Table:

1. INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID , DONER_ID) VALUES (1,10000,TO_DATE('2024-02-15','YYYY-MM-DD'),1,1);

2.INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (2, 20000, TO_DATE('2024-03-20', 'YYYY-MM-DD'), 2, 2);

3.INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (3, 30000, TO_DATE('2024-04-10', 'YYYY-MM-DD'), 3, 3);

4.INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (4, 40000, TO_DATE('2024-04-25', 'YYYY-MM-DD'), 4, 4);

5.INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (5, 50000, TO_DATE('2024-05-05', 'YYYY-MM-DD'), 5, 5);

6.INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (6, 60000, TO_DATE('2024-05-20', 'YYYY-MM-DD'), 6, 6);

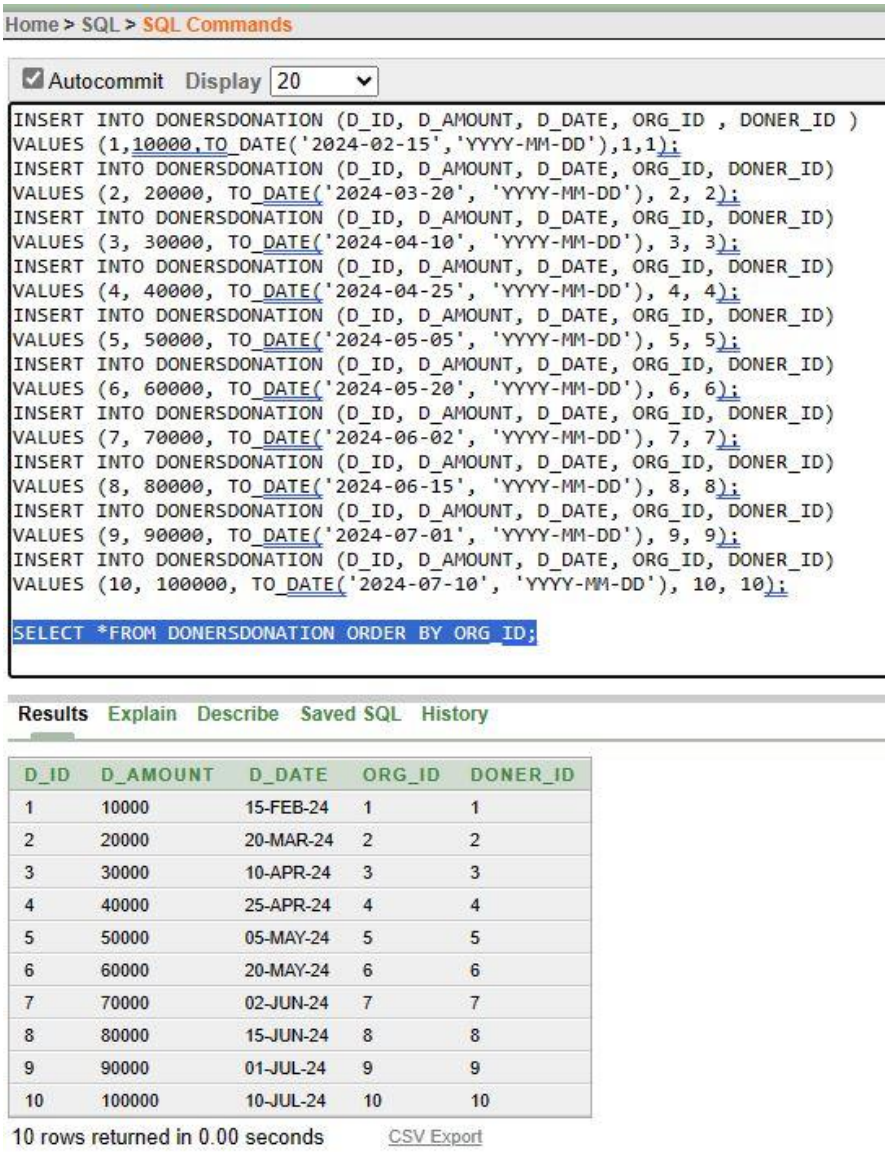
7. INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (7, 70000, TO_DATE('2024-06-02', 'YYYY-MM-DD'), 7, 7);

8. INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (8, 80000, TO_DATE('2024-06-15', 'YYYY-MM-DD'), 8, 8);

9. INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (9, 90000, TO_DATE('2024-07-01', 'YYYY-MM-DD'), 9, 9);

10. INSERT INTO DONERSDONATION (D_ID, D_AMOUNT, D_DATE, ORG_ID, DONER_ID) VALUES (10, 100000, TO_DATE('2024-07-10', 'YYYY-MM-DD'), 10, 10);

SELECT *FROM DONERSDONATION ORDER BY ORG_ID;



Results Explain Describe Saved SQL History

D_ID	D_AMOUNT	D_DATE	ORG_ID	DONER_ID
1	10000	15-FEB-24	1	1
2	20000	20-MAR-24	2	2
3	30000	10-APR-24	3	3
4	40000	25-APR-24	4	4
5	50000	05-MAY-24	5	5
6	60000	20-MAY-24	6	6
7	70000	02-JUN-24	7	7
8	80000	15-JUN-24	8	8
9	90000	01-JUL-24	9	9
10	100000	10-JUL-24	10	10

10 rows returned in 0.00 seconds CSV Export

7.6 Insert into ORGANIZATION Table:

1.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (1, '01856471424', 'CHARITY FOUNDATION', 'DHAKA, BANGLADESH');

2.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (2, '01856471425', 'HEALTHCARE INITIATIVE', 'DHAKA, BANGLADESH');

3.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (3, '01856471426', 'EDUCATION SUPPORT GROUP', 'DHAKA, BANGLADESH');

4.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (4, '01856471427', 'ENVIRONMENTAL NGO', 'DHAKA, BANGLADESH');

5.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (5, '01856471428', 'HUMANITARIAN AID ORG', 'DHAKA, BANGLADESH');

6.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (6, '01856471429', 'COMMUNITY DEVELOPMENT ASSOCIATION', 'DHAKA, BANGLADESH');

7.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (7, '01856471430', 'DISASTER RELIEF AGENCY', 'DHAKA, BANGLADESH');

8.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (8, '01856471431', 'MENTAL HEALTH FOUNDATION', 'DHAKA, BANGLADESH');

9.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (9, '01856471432', 'WILDLIFE CONSERVATION GROUP', 'DHAKA, BANGLADESH');

10.INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS) VALUES (10, '01856471433', 'SUPPORT FOR THE ELDERLY', 'DHAKA, BANGLADESH');

SELECT *FROM ORGANIZATION ORDER BY ORG_ID;

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```
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (1, '01856471424', 'CHARITY FOUNDATION', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (2, '01856471425', 'HEALTHCARE INITIATIVE', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (3, '01856471426', 'EDUCATION SUPPORT GROUP', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (4, '01856471427', 'ENVIRONMENTAL NGO', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (5, '01856471428', 'HUMANITARIAN AID ORG', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (6, '01856471429', 'COMMUNITY DEVELOPMENT ASSOCIATION', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (7, '01856471430', 'DISASTER RELIEF AGENCY', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (8, '01856471431', 'MENTAL HEALTH FOUNDATION', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (9, '01856471432', 'WILDLIFE CONSERVATION GROUP', 'DHAKA, BANGLADESH');
INSERT INTO ORGANIZATION (ORG_ID, ORG_CONTACT, ORG_NAME, ORG_ADDRESS)
VALUES (10, '01856471433', 'SUPPORT FOR THE ELDERLY', 'DHAKA, BANGLADESH');

SELECT *FROM ORGANIZATION ORDER BY ORG_ID;
```

Results Explain Describe Saved SQL History

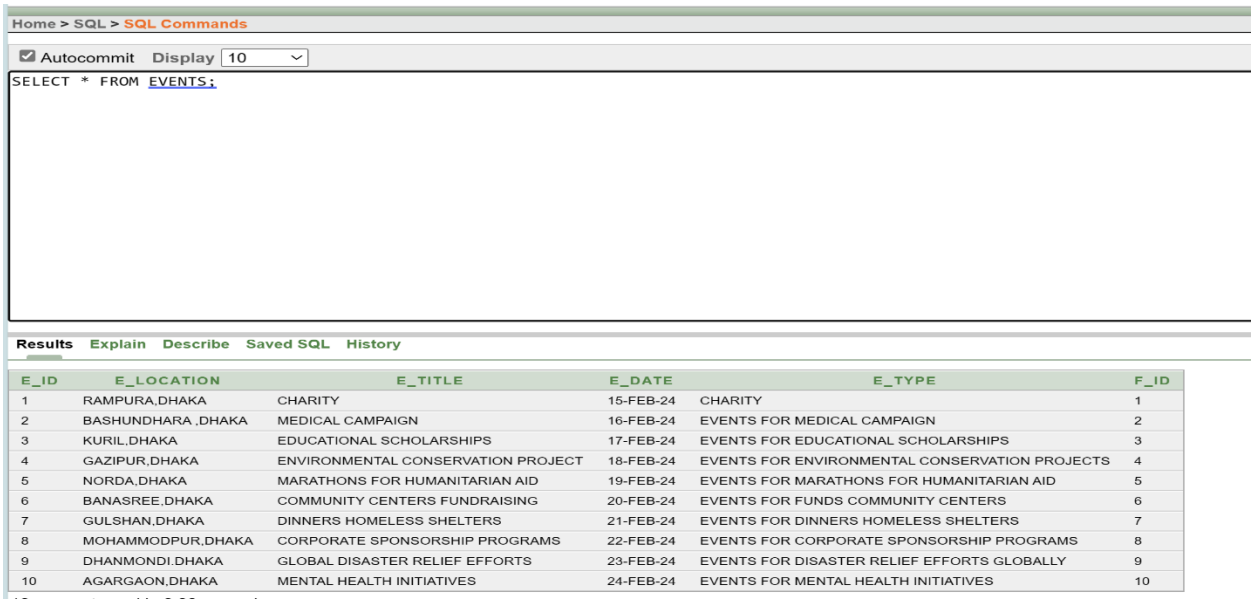
ORG_ID	ORG_CONTACT	ORG_NAME	ORG_ADDRESS
1	01856471424	CHARITY FOUNDATION	DHAKA, BANGLADESH
2	01856471425	HEALTHCARE INITIATIVE	DHAKA, BANGLADESH
3	01856471426	EDUCATION SUPPORT GROUP	DHAKA, BANGLADESH
4	01856471427	ENVIRONMENTAL NGO	DHAKA, BANGLADESH
5	01856471428	HUMANITARIAN AID ORG	DHAKA, BANGLADESH
6	01856471429	COMMUNITY DEVELOPMENT ASSOCIATION	DHAKA, BANGLADESH
7	01856471430	DISASTER RELIEF AGENCY	DHAKA, BANGLADESH
8	01856471431	MENTAL HEALTH FOUNDATION	DHAKA, BANGLADESH
9	01856471432	WILDLIFE CONSERVATION GROUP	DHAKA, BANGLADESH
10	01856471433	SUPPORT FOR THE ELDERLY	DHAKA, BANGLADESH

10 rows returned in 0.00 seconds CSV Export

8. Query

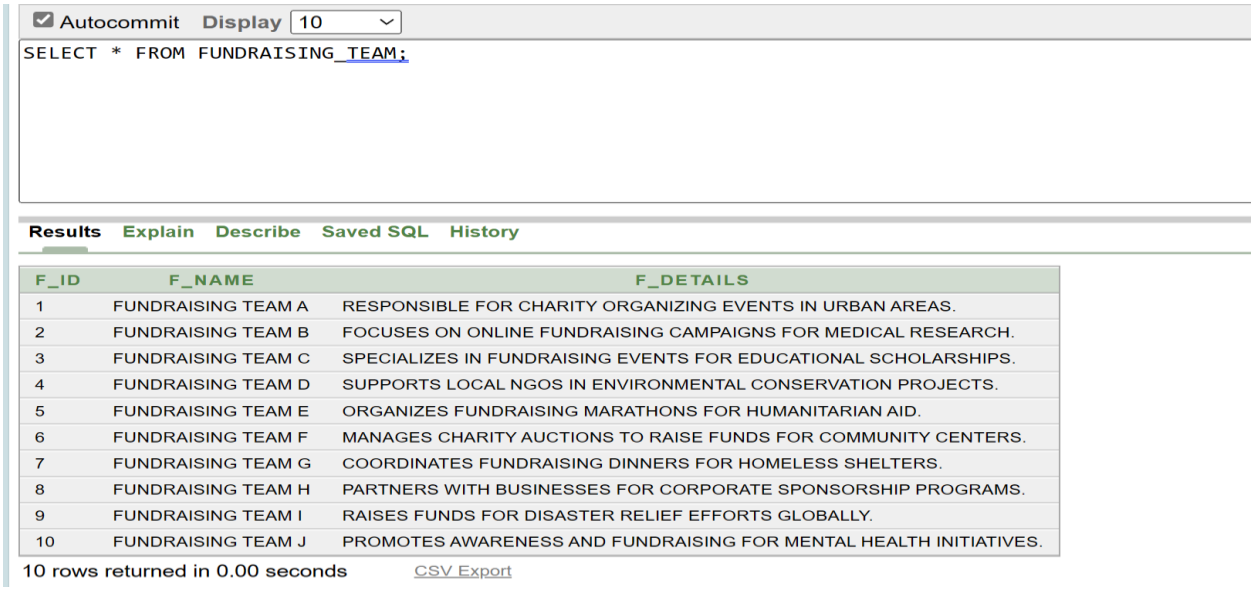
8.1 Basic Query (4 Queries):

1. SELECT * FROM EVENTS;



E_ID	E_LOCATION	E_TITLE	E_DATE	E_TYPE	F_ID
1	RAMPURA,DHAKA	CHARITY	15-FEB-24	CHARITY	1
2	BASHUNDHARA ,DHAKA	MEDICAL CAMPAIGN	16-FEB-24	EVENTS FOR MEDICAL CAMPAIGN	2
3	KURIL,DHAKA	EDUCATIONAL SCHOLARSHIPS	17-FEB-24	EVENTS FOR EDUCATIONAL SCHOLARSHIPS	3
4	GAZIPUR,DHAKA	ENVIRONMENTAL CONSERVATION PROJECT	18-FEB-24	EVENTS FOR ENVIRONMENTAL CONSERVATION PROJECTS	4
5	NORDA,DHAKA	MARATHONS FOR HUMANITARIAN AID	19-FEB-24	EVENTS FOR MARATHONS FOR HUMANITARIAN AID	5
6	BANASREE,DHAKA	COMMUNITY CENTERS FUNDRAISING	20-FEB-24	EVENTS FOR FUNDS COMMUNITY CENTERS	6
7	GULSHAN,DHAKA	DINNERS HOMELESS SHELTERS	21-FEB-24	EVENTS FOR DINNERS HOMELESS SHELTERS	7
8	MOHAMMODPUR,DHAKA	CORPORATE SPONSORSHIP PROGRAMS	22-FEB-24	EVENTS FOR CORPORATE SPONSORSHIP PROGRAMS	8
9	DHANMONDI.DHAKA	GLOBAL DISASTER RELIEF EFFORTS	23-FEB-24	EVENTS FOR DISASTER RELIEF EFFORTS GLOBALLY	9
10	AGARGAON,DHAKA	MENTAL HEALTH INITIATIVES	24-FEB-24	EVENTS FOR MENTAL HEALTH INITIATIVES	10

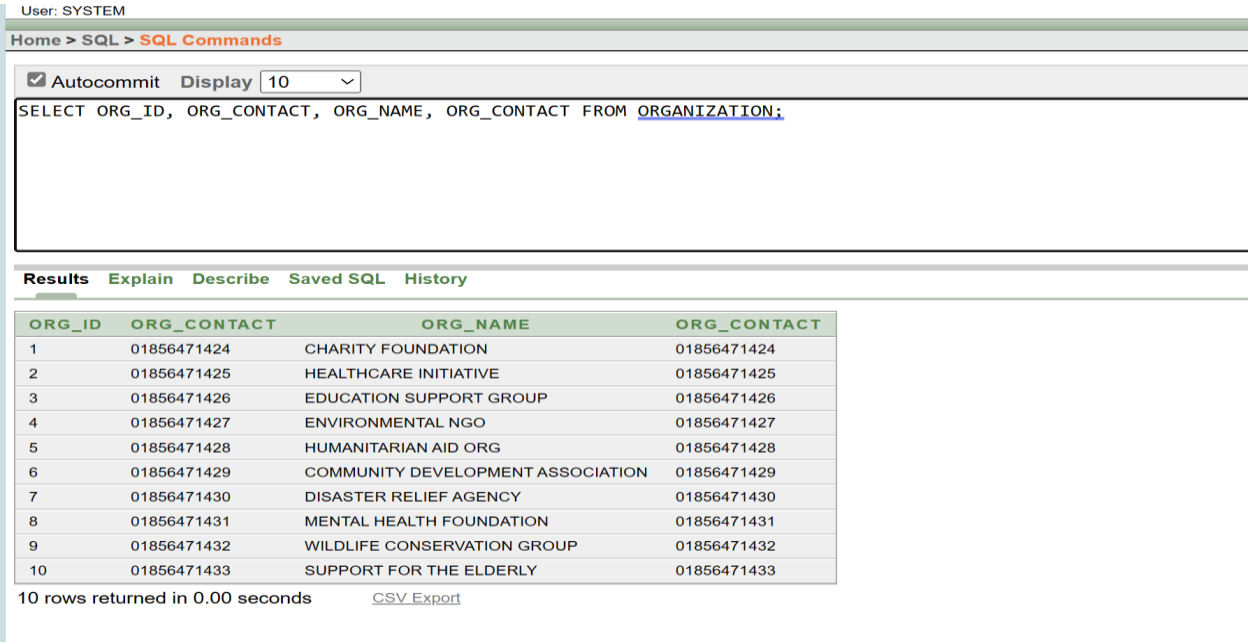
2. SELECT * FROM FUNDRAISING_TEAM;



F_ID	F_NAME	F_DETAILS
1	FUNDRAISING TEAM A	RESPONSIBLE FOR CHARITY ORGANIZING EVENTS IN URBAN AREAS.
2	FUNDRAISING TEAM B	FOCUSES ON ONLINE FUNDRAISING CAMPAIGNS FOR MEDICAL RESEARCH.
3	FUNDRAISING TEAM C	SPECIALIZES IN FUNDRAISING EVENTS FOR EDUCATIONAL SCHOLARSHIPS.
4	FUNDRAISING TEAM D	SUPPORTS LOCAL NGOS IN ENVIRONMENTAL CONSERVATION PROJECTS.
5	FUNDRAISING TEAM E	ORGANIZES FUNDRAISING MARATHONS FOR HUMANITARIAN AID.
6	FUNDRAISING TEAM F	MANAGES CHARITY AUCTIONS TO RAISE FUNDS FOR COMMUNITY CENTERS.
7	FUNDRAISING TEAM G	COORDINATES FUNDRAISING DINNERS FOR HOMELESS SHELTERS.
8	FUNDRAISING TEAM H	PARTNERS WITH BUSINESSES FOR CORPORATE SPONSORSHIP PROGRAMS.
9	FUNDRAISING TEAM I	RAISES FUNDS FOR DISASTER RELIEF EFFORTS GLOBALLY.
10	FUNDRAISING TEAM J	PROMOTES AWARENESS AND FUNDRAISING FOR MENTAL HEALTH INITIATIVES.

10 rows returned in 0.00 seconds CSV Export

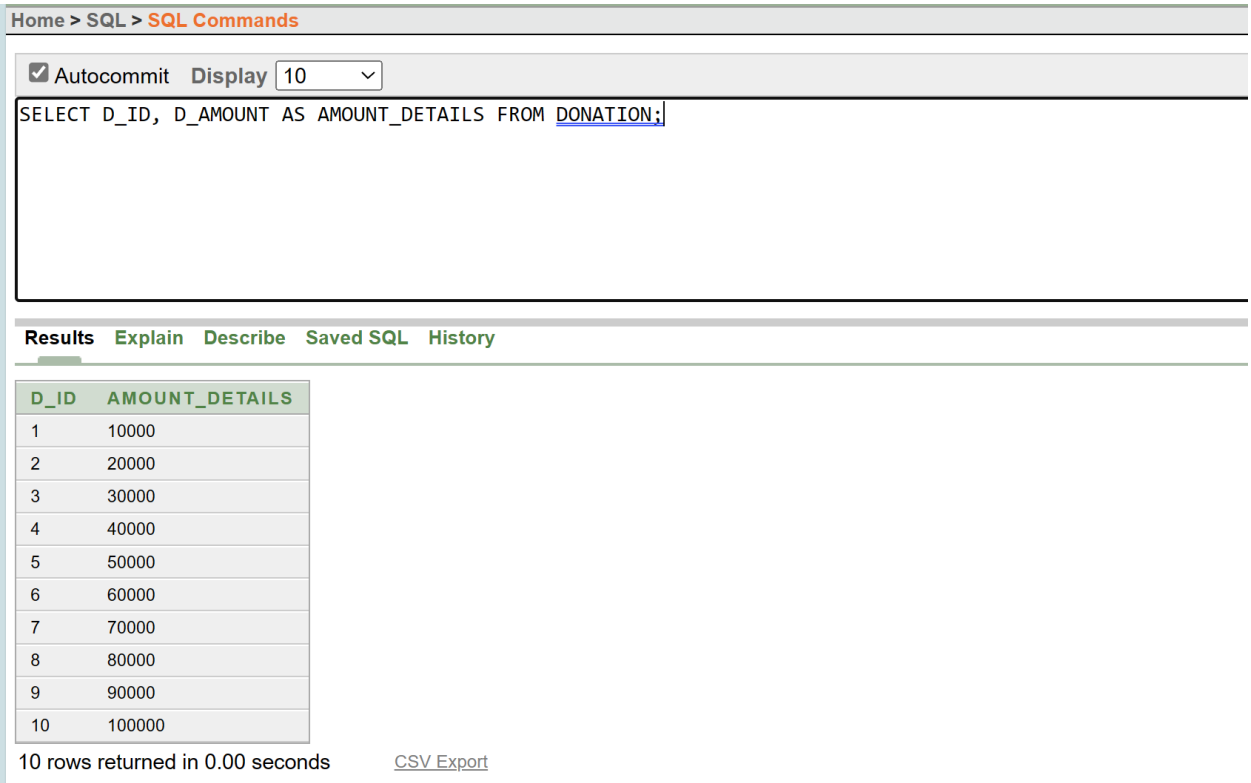
3. SELECT ORG_ID, ORG_CONTACT, ORG_NAME, ORG_CONTACT FROM ORGANIZATION;



ORG_ID	ORG_CONTACT	ORG_NAME	ORG_CONTACT
1	01856471424	CHARITY FOUNDATION	01856471424
2	01856471425	HEALTHCARE INITIATIVE	01856471425
3	01856471426	EDUCATION SUPPORT GROUP	01856471426
4	01856471427	ENVIRONMENTAL NGO	01856471427
5	01856471428	HUMANITARIAN AID ORG	01856471428
6	01856471429	COMMUNITY DEVELOPMENT ASSOCIATION	01856471429
7	01856471430	DISASTER RELIEF AGENCY	01856471430
8	01856471431	MENTAL HEALTH FOUNDATION	01856471431
9	01856471432	WILDLIFE CONSERVATION GROUP	01856471432
10	01856471433	SUPPORT FOR THE ELDERLY	01856471433

10 rows returned in 0.00 seconds CSV Export

4. SELECT D_ID, D_AMOUNT AS AMOUNT_DETAILS FROM DONATION;

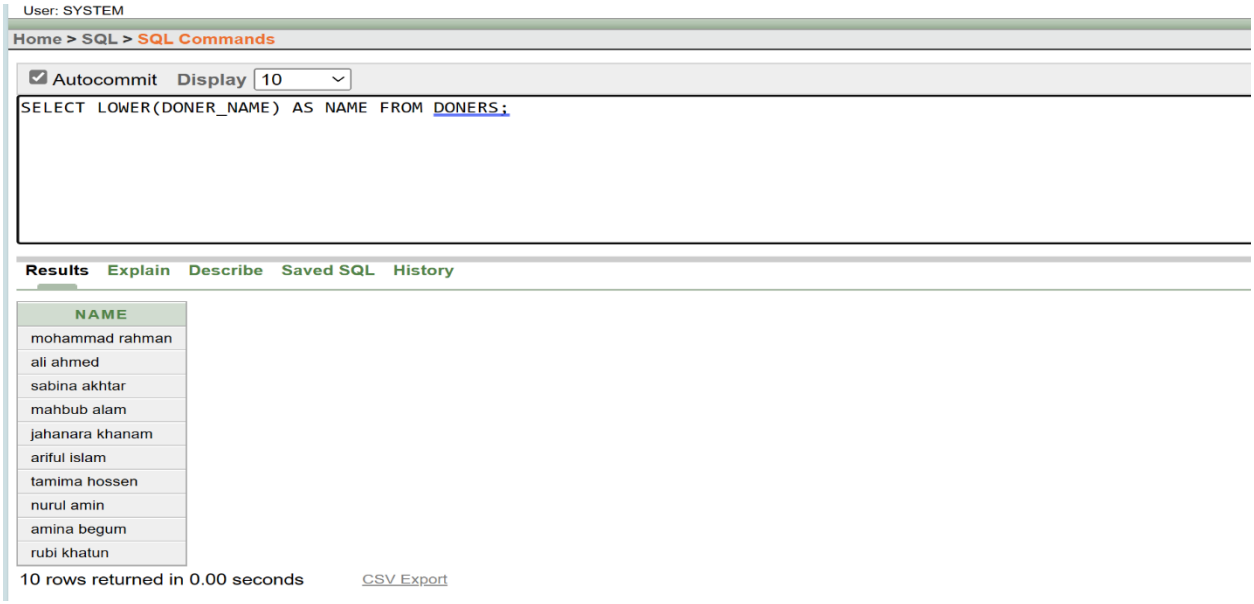


D_ID	AMOUNT_DETAILS
1	10000
2	20000
3	30000
4	40000
5	50000
6	60000
7	70000
8	80000
9	90000
10	100000

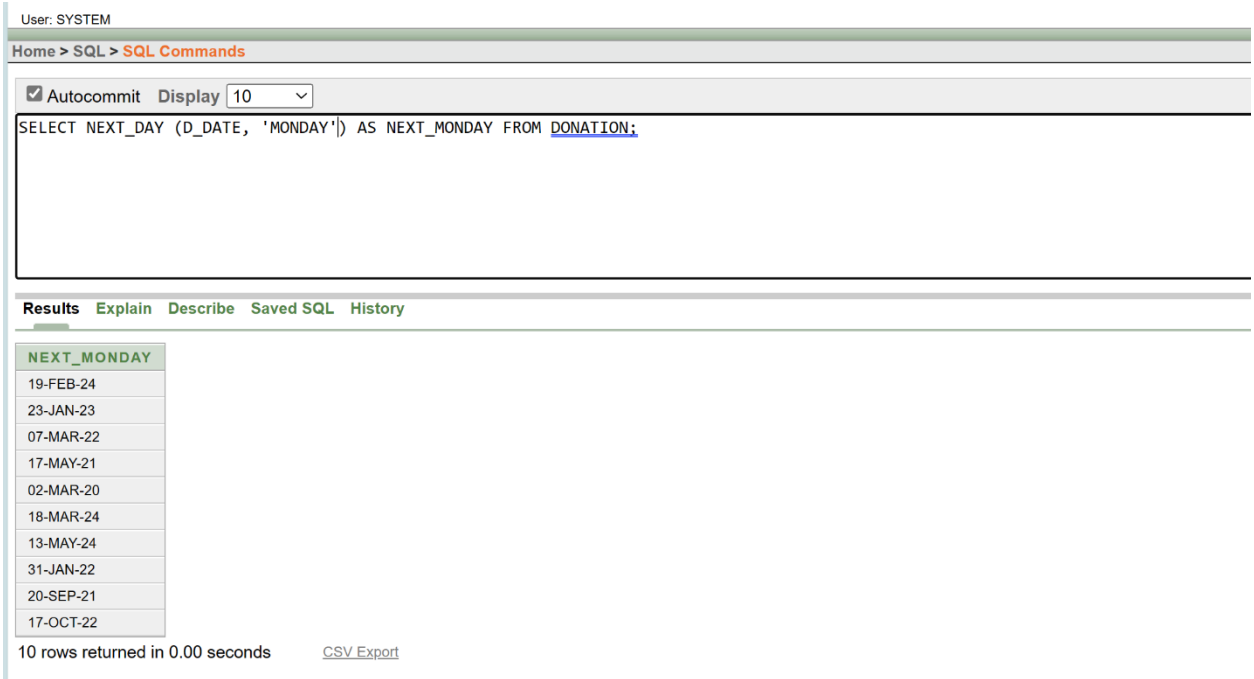
10 rows returned in 0.00 seconds CSV Export

MAKING TWO QUERIES USING SINGLE ROW FUNCTION:

1. SELECT LOWER(DONER_NAME) AS NAME FROM DONERS;

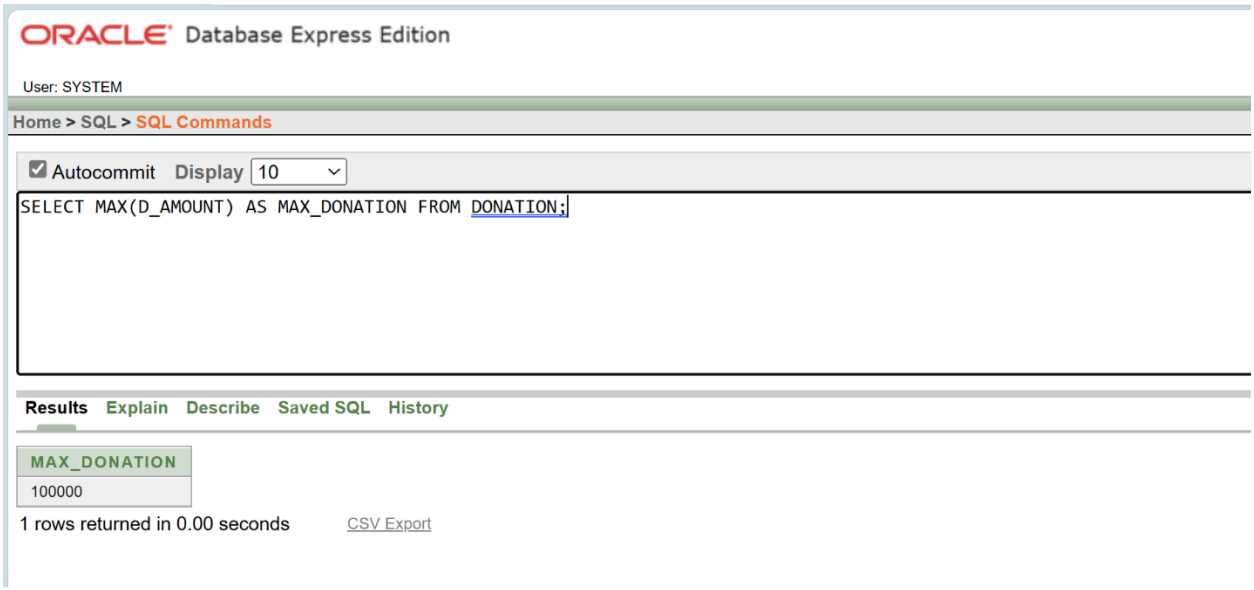


2. SELECT NEXT_DAY (D_DATE, 'MONDAY') AS NEXT_MONDAY FROM DONATION;

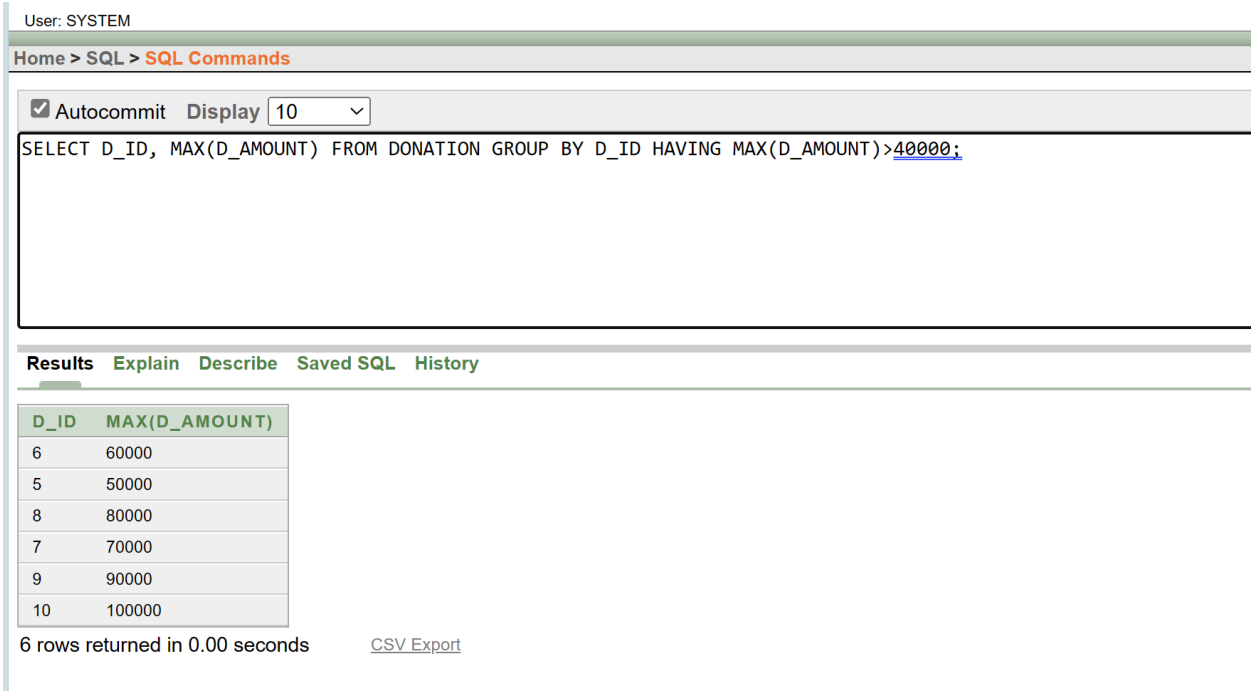


MAKING TWO QUERIES USING GROUP FUNCTION:

1. SELECT MAX(D_AMOUNT) AS MAX_DONATION FROM DONATION;

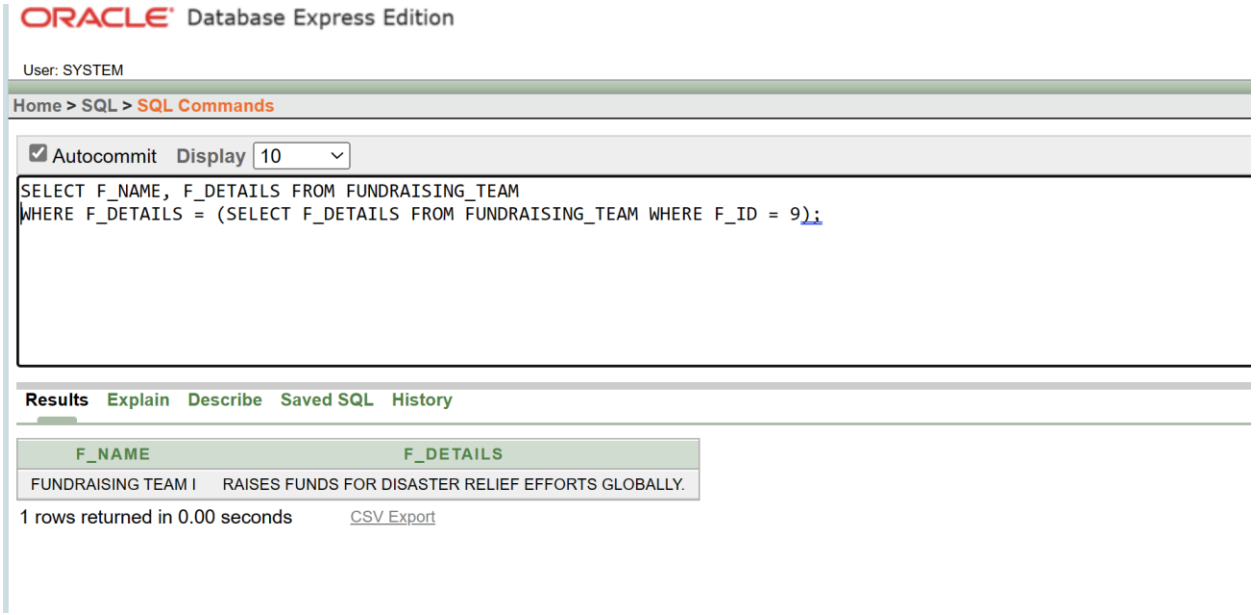


2. SELECT D_ID, MAX(D_AMOUNT) FROM DONATION GROUP BY D_ID HAVING MAX(D_AMOUNT)>40000;

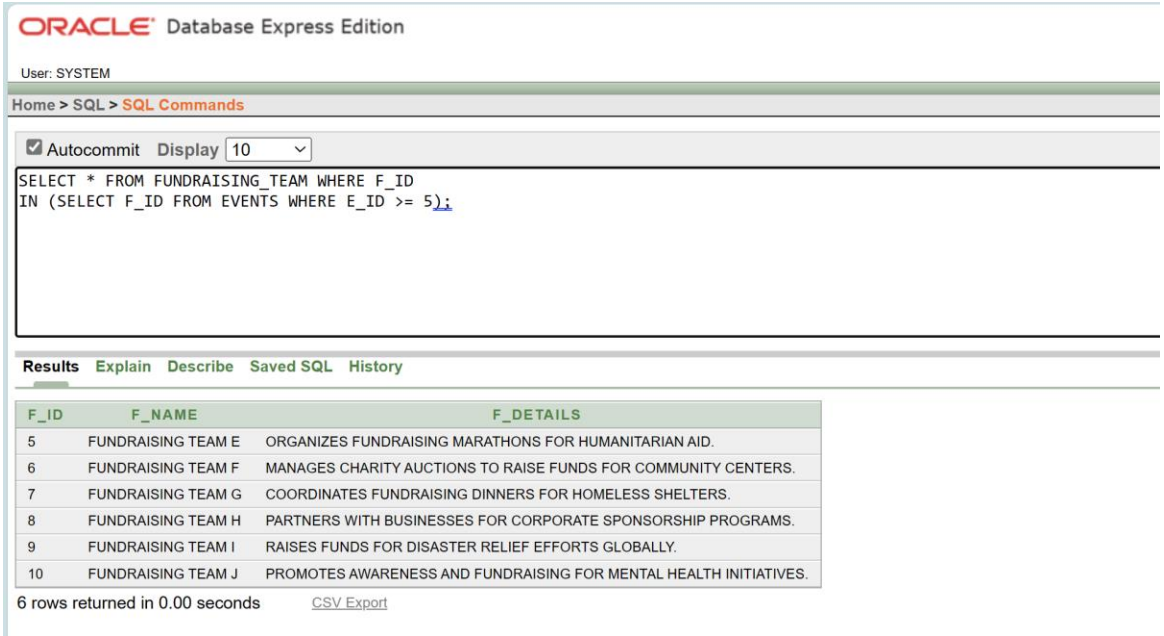


MAKING TWO QUERIES USING SUBQUERY:

1. SELECT F_NAME, F_DETAILS FROM FUNDRAISING_TEAM WHERE F_DETAILS = (SELECT F_DETAILS FROM FUNDRAISING_TEAM WHERE F_ID = 9);



2. SELECT * FROM FUNDRAISING_TEAM WHERE F_ID IN (SELECT F_ID FROM EVENTS WHERE E_ID >= 5);

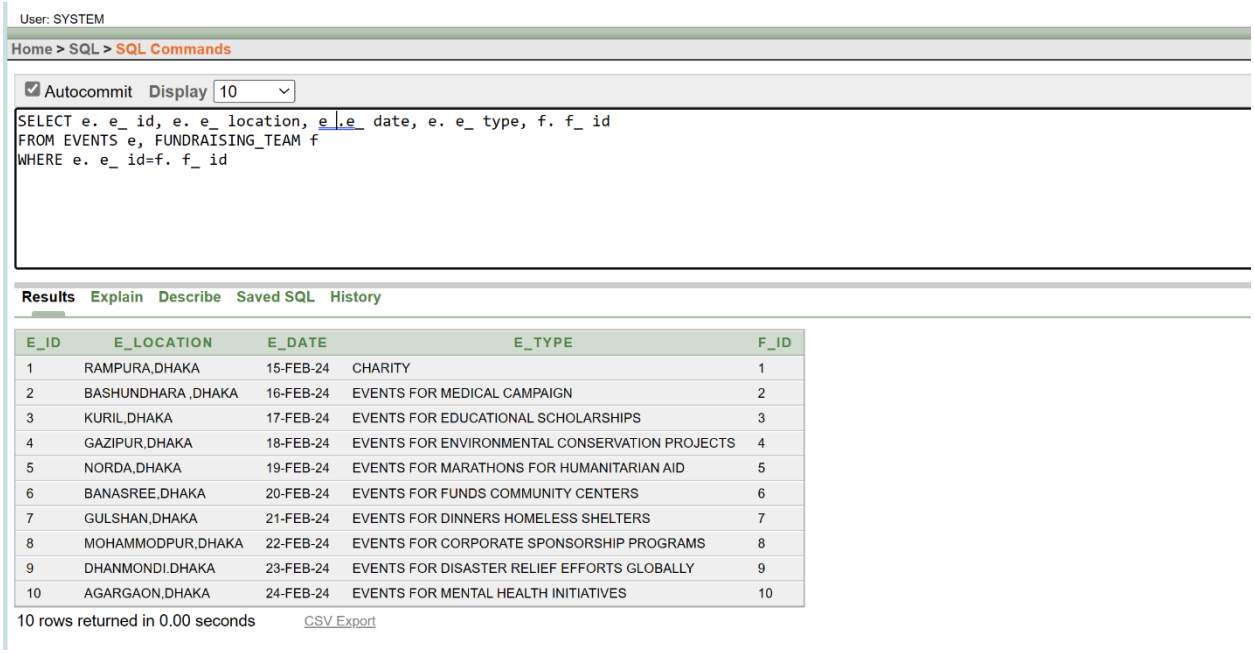


MAKING TWO QUERIES USING JOINING

1. SELECT e.e_id, e.e_location,e.e_date, e.e_type, f.f_id

FROM EVENTS e,FUNDRAISING_TEAM f

WHERE e.e_id=f.f_id



E_ID	E_LOCATION	E_DATE	E_TYPE	F_ID
1	RAMPURA,DHAKA	15-FEB-24	CHARITY	1
2	BASHUNDHARA ,DHAKA	16-FEB-24	EVENTS FOR MEDICAL CAMPAIGN	2
3	KURIL,DHAKA	17-FEB-24	EVENTS FOR EDUCATIONAL SCHOLARSHIPS	3
4	GAZIPUR,DHAKA	18-FEB-24	EVENTS FOR ENVIRONMENTAL CONSERVATION PROJECTS	4
5	NORDA,DHAKA	19-FEB-24	EVENTS FOR MARATHONS FOR HUMANITARIAN AID	5
6	BANASREE,DHAKA	20-FEB-24	EVENTS FOR FUNDS COMMUNITY CENTERS	6
7	GULSHAN,DHAKA	21-FEB-24	EVENTS FOR DINNERS HOMELESS SHELTERS	7
8	MOHAMMODPUR,DHAKA	22-FEB-24	EVENTS FOR CORPORATE SPONSORSHIP PROGRAMS	8
9	DHANMONDI.DHAKA	23-FEB-24	EVENTS FOR DISASTER RELIEF EFFORTS GLOBALLY	9
10	AGARGAON,DHAKA	24-FEB-24	EVENTS FOR MENTAL HEALTH INITIATIVES	10

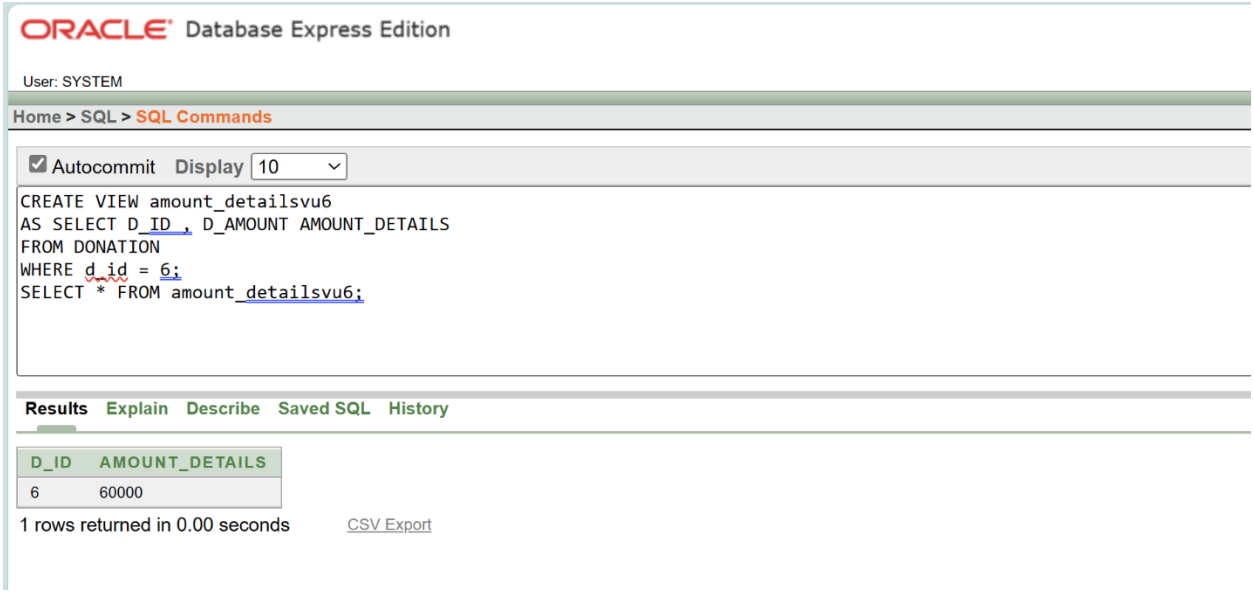
10 rows returned in 0.00 seconds CSV Export

2.

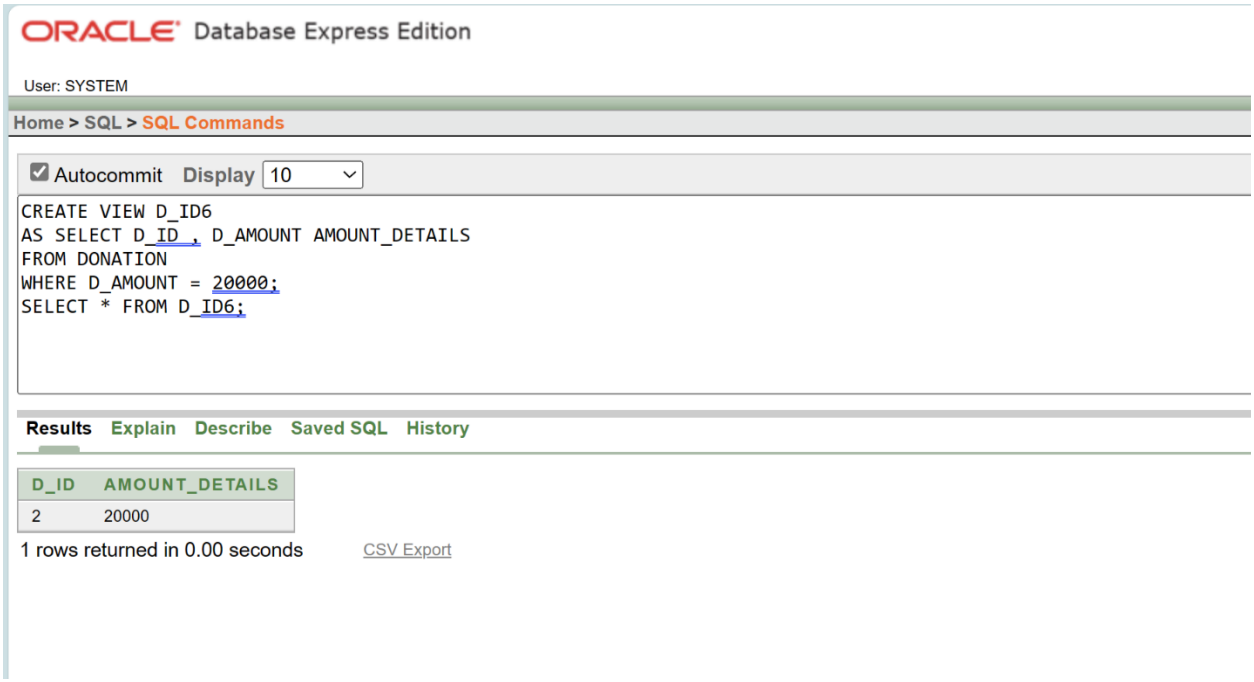
MAKE TWO QUERIES USING VIEW

1. CREATE VIEW amount_detailsvu6 AS SELECT D_ID , D_AMOUNT AMOUNT_DETAILS

FROM DONATIONWHERE d_id = 6; SELECT * FROM amount_detailsvu6;



2.



9.RELATIONAL ALGEBRA:

1 Projecting specific columns from EVENTS table:

$$R1 = \pi_{E_TITLE,E_LOCATION,E_DATE}(EVENIS)$$

Explanation: This query selects only the columns 'E_TITLE', 'E_LOCATION', and 'E_DATE' from the 'EVENTS' table.

2. Joining FUNDRAISING_TEAM and EVENTS tables and selecting specific columns:

$$R2 = \pi_{F_NAME,E_TITLE}(FUNDRAISING_TEAM \bowtie EVENTS)$$

Explanation:

This query joins the 'FUNDRAISTNG_TEAM' and 'EVENTS' tables based on the foreign key relationship and then selects columns 'F_NAME' from 'FUNDRAISING_TEAM' and 'E_TITLE' from 'EVENTS'.

3. Joining DONERSDONATION and DONERS tables and selecting specific columns:

$$R3 = \pi_{DONER_NAME,D_AMOUNT}(DONERSDONATION \bowtie DONERS)$$

Explanation: This query joins the 'DONERSDONATION' and 'DONERS' tables based on the foreign key relationship and then selects columns 'DONER_NAME' from 'DONERS' and 'D_AMOUNT' from 'DONERSDONATION".

4. Joining EVENTS and DONATION tables, then projecting and renaming columns:

$$R4 = \rho(EVENT_TITLB: E_TITLE)\left(\sigma_{B_DATE>2024-03-01}(EVENIS \propto DONATION)\right)$$

Explanation:

This query first joins the 'EVENTS' and 'DONATION' tables based on the foreign key relationship, then selects events with a date after "2024-03-0r', and finally renames the 'E_TITLE' column to 'EVENT_TRTLE'.

5. Joining ORGANIZATION, DONERSDONATION, and DONATION tables, then projecting specific columns:

$$R5 = \pi O_{NAME,D_AMOUNI}(ORGANIZATION \bowtie (DONERSDONATION \bowtie DO$$

Explanation: This query joins the 'OREANIZATION', 'DONERSDONATION', and 'DONATION' tables based on their foreign key relationships, then selects columns 'ORG_NAME' from 'ORCANTZATION* and 'D_AMOUNT' from the joined tables.

10. CONCLUSION:

The Fundraiser Management System (FMS) is an innovative solution designed to simplify and optimize fundraising campaigns for individuals, charities, and organizations. Its core purpose revolves around streamlining the complexities inherent in organizing and executing successful fundraising initiatives. FMS stands out for its comprehensive suite of tools and features, offering users an intuitive interface to manage campaigns efficiently. Through FMS, users can set goals, define strategies, and track progress seamlessly, making campaign management a streamlined process. One of the system's key strengths lies in its ability to centralize various aspects of fundraising, including DONER management, online fundraising, event planning, and reporting. This centralized approach fosters greater coordination and effectiveness in fundraising efforts. Additionally, FMS incorporates data analytics and engagement tools, empowering users to cultivate meaningful relationships with DONERs and optimize resource allocation. Overall, FMS plays a pivotal role in empowering organizations to navigate fundraising intricacies with confidence, driving greater efficiency, effectiveness, and ultimately, making a tangible difference in their communities.